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PALAEOMAGNETISM OF UPPER CRETACEOUS VOLCANIC ROCKS FROM CABO DE STO. AGOSTINHO, BRAZIL

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One hundred samples from nine sites in Upper Cretaceous volcanics (K/Ar age 85–99 m.y.) of the magmatic province of Cabo de Santo Agostinho, Pernambuco (8.4°S, 35.0°W) yield a mean direction of magnetization $D = 0.4°$, $I = -20.6°$ with $\alpha_{95} = 4.8°$, $k = 114$ after AF cleaning. All sites have normal polarity with a mean pole, named SAK_{10}, at 87.6°N, 135°E with $A_{95} = 4.5°$ which is close to other Upper Cretaceous poles for South America. These poles are compared with Upper Cretaceous poles of Africa for various reconstructions of the two continents.

1. Introduction

The palaeomagnetism of Upper Cretaceous rocks from South America is of interest because of its relevance to the opening of the South Atlantic. Until now, few Upper Cretaceous poles for South America have been published [1,2]. In this paper a new pole position from northeast Brazil is presented.

2. Geological setting

About 30 km south of Recife, Pernambuco, several types of magmatic rocks varying from basaltic to granite crop out (Fig. 1). The magmatic rocks of Cabo de Santo Agostinho cut through Cretaceous sediments of the Cabo Formation [3]. Some of the volcanic rocks have been dated by the K/Ar method indicating an Upper Cretaceous age [4]: trachyte (at site 42, Fig. 1) 85 and 89 m.y. and rhyolite (at site 37) 94 and 99 m.y. The latter is believed to be cogenetic with the granite of Cabo de Sto. Agostinho with an age of about 90 m.y. [3].

3. Sampling, measurements, and results

The samples (2.5 cm in diameter) were cored with a portable drilling machine. Locations of the 11 sites sampled are shown in Fig. 1. From each site about ten samples were taken. The natural remanent magnetizations (NRM) of all samples were measured with a "Digico" spinner magnetometer. Pilot samples of all sites were then demagnetized in steps up to 1000 Oe AF peak values and, according to these results, all samples were treated with the appropriate alternating field (Table 1). The site mean directions of NRM and of the characteristic remanence (CARM) are shown in Fig. 2, and in Table 1 the palaeomagnetic results are summarized. The within-site scatter, as well as the between-site scatter, could be reduced by the AF treatment. There is a wide range of medium destruc-

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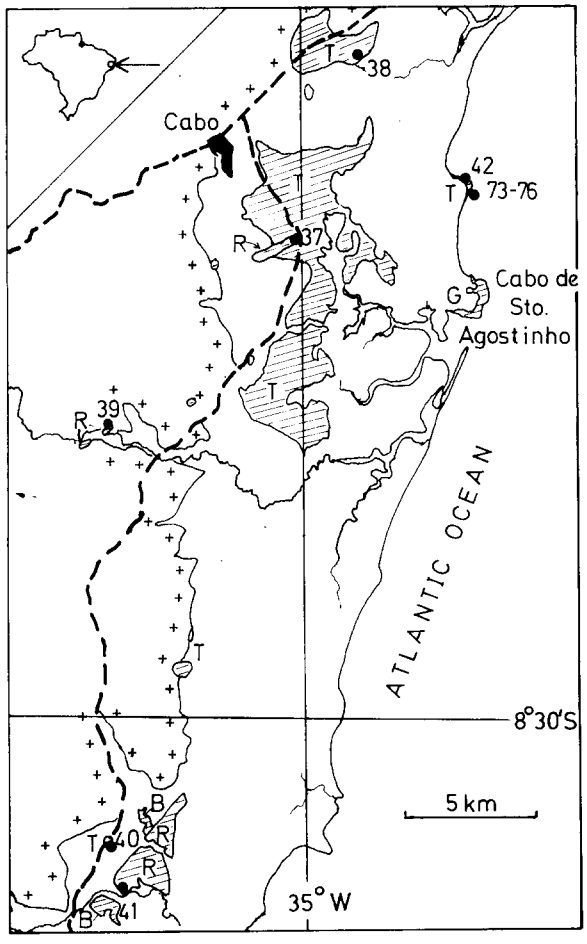


Fig. 1. Geological sketch map of the magmatic province of Cabo de Santo Agostinho, Pernambuco [5], showing sampling sites. Hatched areas denote volcanic rocks: *R* = rhyolite, *T* = trachyte, *B* = basalt, *G* = granite; ++ = basement rocks.

tive field (Table 1) from 30 to about 900 Oe and, correspondingly, the intensity of NRM which varied between 10^{-1} and 1 A m^{-1} was reduced to about 80 to 5% of its initial value by the applied alternating field. For neighbouring sites (39/1, 39/6 and 42, 73-76) an *F*-test [6] was applied to compare the directions of CARM. Only those with significantly dif-

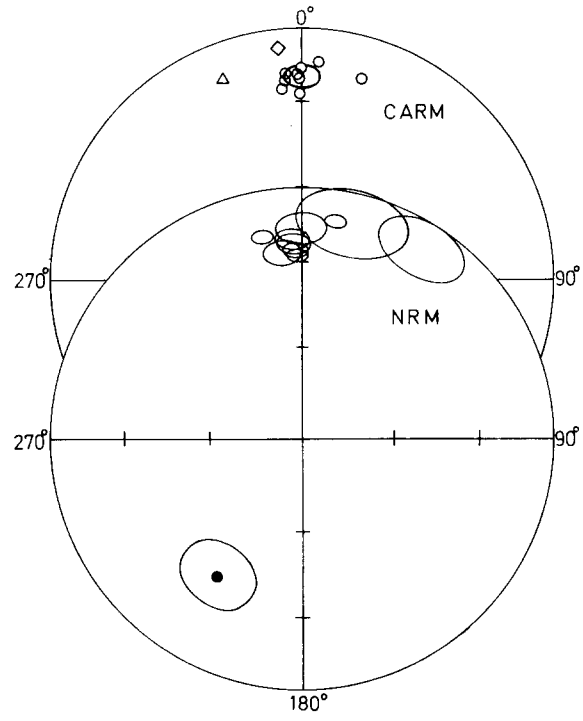


Fig. 2. Equal area projection of site means. For the NRM directions circles of 95% confidence are shown. For CARM only the site means (small circles) and the circle of confidence for the overall mean (with the exception of one the radii of the circles of confidence for CARM site means are smaller than 4.5°; see Table 1). The diamond and triangle denote the present central dipole field direction and the actual earth's magnetic field direction, respectively. Open symbols for negative inclination, closed symbols for positive inclination.

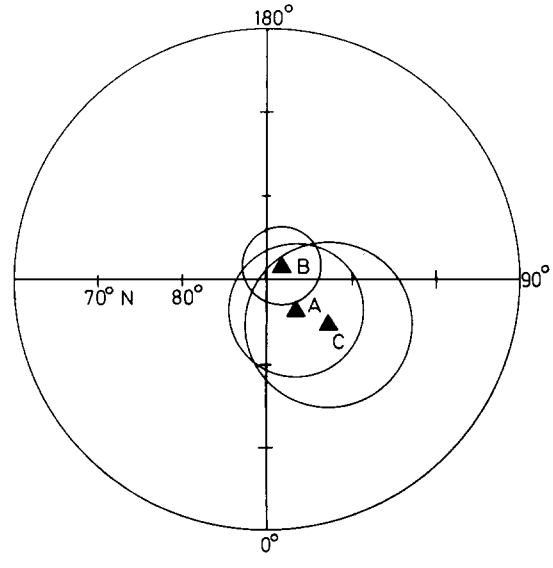


Fig. 3. Mean Upper Cretaceous poles with circles of confidence for South America. *A* = Pirgua (SAK_6); *B* = Cabo de Santo Agostinho (SAK_{10}); *C* = Poços de Caldas (SAK_7). See Table 2 for details.

TABLE 1

Palaeomagnetic results for volcanic rocks from Cabo de Santo Agostinho (mean site location 8.4°S, 35.0°W)

Site	NRM					MDF (Oe)	AF (Oe)	CARM					VGP	
	N (°)	D (°)	I (°)	α_{95} (°)	k			N (°)	D (°)	I (°)	α_{95} (°)	k	°N	°E
37	7	357.7	−23.5	4.4	183	100	120	9	354.9	−22.1	4.0	163	84.1	203
38	10	358.5	−25.8	3.2	229	330	400	10	358.8	−19.8	2.7	320	87.8	177
39/1	5	211.5	+35.9	12.2	40	30	400	6	354.1	−24.8	4.5	220	82.6	195
39/6	5	32.7	−11.2	11.2	48	30	400	5	16.9	−18.0	12.0	41	73.3	59
40	11	353.7	−26.9	4.9	86	60	75	10	355.4	−19.1	2.1	493	85.3	217
41	10	359.4	−27.9	1.4	1110	120	130	13	359.4	−27.1	1.1	1300	83.9	150
42	10	9 0	−13.6	2.7	313	900	300	10	4.8	−14.7	2.6	340	85.2	42
73	11	0.0	−17.5	6.3	57	120	200							
74	11	349.1	−19.8	2.7	270	200	300	30	359.3	−21.3	1.3	378	87.2	158
76	9	356.7	−21.6	3.9	169	900	400							
75	7	13.3	−12.4	14.6	18	180	300	7	359.9	−17.4	3.7	258	89.4	149
								9	0.4	−20.6	4.8	114	87.6	135.1
													A_{95} = 4.5	K = 127

N = number of samples, D = declination, I = inclination, α_{95} or A_{95} = radius of 95% confidence circle, k or K = precision parameter, MDF = medium destructive peak field necessary to erase half the NMR intensity by alternating field demagnetization, AF = peak value of alternating field used for demagnetization (10^4 Oe = 1 T).

TABLE 2

Upper Cretaceous palaeomagnetic poles for South America and Africa

Label	Formation (name)	Age (m.y.)	Pole		A_{95} (°)	110 m.y. B.P. [1]		95 m.y. [2]		80 m.y. [3]		References
			°N	°E		°N	°E	°N	°E	°N	°E	
A	Pirgua (SAK_6)	77–114	85	42	8	64	256	73	252	82	241	[2]
B	Cabo de Sto. Agostinho (SAK_{10})	85– 99	88	135	4.5	59	252	68	247	77	235	this paper
C	Poços de Caldas (SAK_7)	75– 80	81	53	10	68	252	77	243	85	211	[1]
1	Lupata volcanics	106–111	62	260	3.5							[8]
2	Wadi Natash volcanics	75–100	69	262	5.7							[9]
3	Kimberlite pipes *	83– 89	58	235	4.1							[10]
4	Red siltstone	Ku	79	208	6							[8]
5	Nubian sandstone	Ku	79	220	4.1							[11]
6	Tororo	Ku (?)	76	195	9							[12]
7	Volcanics, Sicily	71– 81	63	229	3.8							[13]

Parameters (poles and angles of rotation) for rotated South American pole positions (South America with respect to Africa) [14]: [1] 49.2°N, 31.8°W, 48.3°; [2] 56.6°N, 34.1°W, 40.1°; [3] 67.3°N, 39.5°W, 32.8°.

* Alternative calculation of mean gives pole at 58°N, 237°E with A_{95} = 15° [10].

ferent directions at the 95% probability level were considered as individual sites: otherwise they were considered as one site (e.g. 73, 74, 76; Table 1). This procedure yields nine individual sites, all with a normal direction of magnetization. Giving unit weight to each site the mean direction of CARM ($D = 0.4°$, $I = -20.6°$ with $\alpha_{95} = 4.8°$, $k = 114$, $N = 9$) is significantly different from the present earth's field (Fig. 2). The angular standard deviation ($s \approx 81/\sqrt{k}$) equals 7.6 and is relatively small. The pole position (135°E, 87.6°N with $A_{95} = 4.5°$) is not significantly different from other known Upper Cretaceous poles for South America (Fig. 3, Table 2) but it does differ from Lower Cretaceous pole positions (e.g. [7]). The new Upper Cretaceous pole has been named SAK_{10}.

4. Discussion

In Table 2, Upper Cretaceous poles for South America and Africa are compiled. These pole positions may be compared for reconstructions of the two continents for the Upper Cretaceous. The respective "rotated" poles of South America (Africa in its present position) are given in Table 2 for the 110-, 95- and 80-m.y. B.P. reconstructions [14]. It can be seen in Fig. 4 that the 95-m.y. reconstruction brings the Cabo de Sto. Agostinho pole near to a newly determined Upper Cretaceous pole for the Wadi Natash volcanics, Egypt [9]. Both formations have similar radiometric ages, and for both formations only normal magnetization directions were determined. The ages are therefore consistent with the long normal polarity sequence from about 110 to 78 m.y. B.P. [15]. The pole of younger age from Poços de Caldas [1] may be compared with the pole from the Nubian sandstone [11]. The majority of the sites (12 out of 18) of the Nubian sandstone have reversed polarity indicating an age younger than 78 m.y. B.P. (younger than the long normal polarity sequence in the Upper Cretaceous [15]). This is also the case for the Poços de Caldas Formation (6 out of 7 sites have reversed polarity) consistent with its radiometric age. These two poles and two other Upper Cretaceous poles from Africa do not differ significantly for the 80-m.y. reconstruction (Fig. 4). The wide range of age of the Pirgua formation (Table 2) allows us to test reconstructions for 80, 95 and 110 m.y. B.P., each

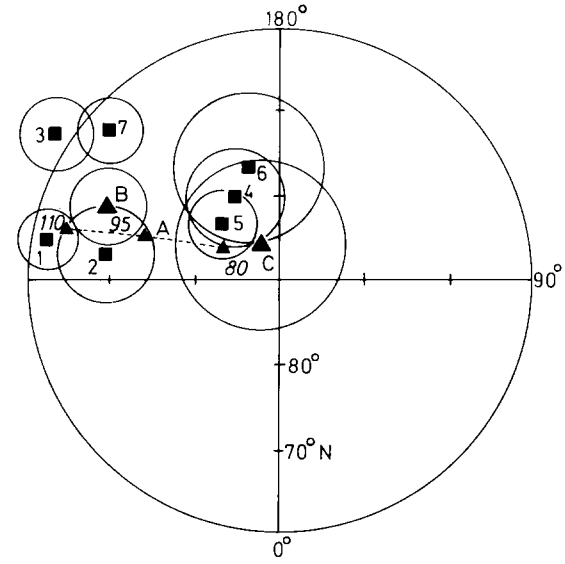


Fig. 4. Comparison of South American (triangles) and African (rectangles) Upper Cretaceous poles with their respective circles of confidence for reconstructions for 80, 95, and 110 m.y. B.P. (marked in italics) [14]. African poles in their present positions. Labels refer to Table 2.

one with a satisfactory result (see Fig. 4). The situation in Fig. 4 does not change significantly if we use other rotation parameters (e.g. [16]) yielding slightly different reconstructions.

There is no agreement between the rotated South American poles and the poles from Sicily and the Kimberlite pipes (Fig. 4, Table 2). It may be argued [9] that the Sicilian pole does not belong to the African polar wander path, and for the Kimberlite pole an alternative calculation of the mean assuming fewer individual sites (Table 2) yields a relatively large A_{95} [10]. These uncertainties suggest that for the Upper Cretaceous which spans about 35 m.y., the number of poles as yet determined for both continents is not sufficiently great.

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